

* Decision Tree Regressor:

Data set

Exp

2

2.5

3

4

4.5

Gap

Yes

Yes

No

No

Yes

Salary

40K

42K

52K

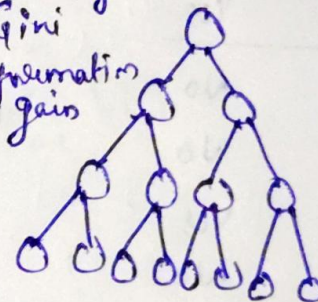
60K

56K

50K

Entropy
Gini
Information
gain

CART



Variance Reduction (Regression Problem)

$$1. \text{ Variance} = \frac{1}{n} \sum_{i=1}^n (y - \bar{y})^2 \quad \text{[mean squared error]}$$

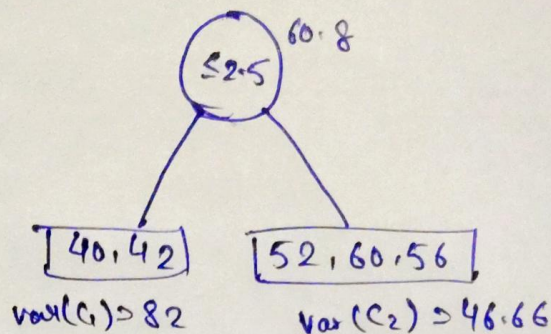
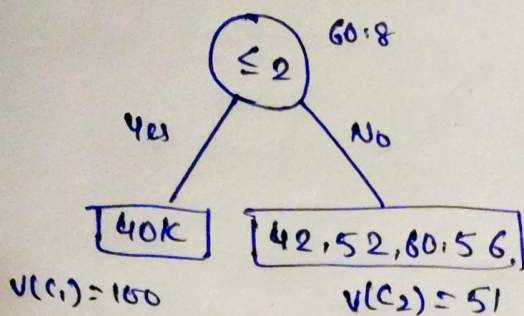
Average

$$\text{Variance (Root)} = \frac{1}{5} [(40 - 50)^2 + (42 - 50)^2 + (52 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$$

$$\boxed{\text{Variance (Root)} = 60.8}$$

40K, 42K, 52K, 60K, 56K

40K, 42K, 52K, 60K, 56K



$$\text{Variance}(C_1) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$= \frac{1}{1} (40 - 50)^2$$

$$= 100$$

$$\text{Variance}(C_2) = \frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2$$

$$= \frac{1}{4} [(42 - 50)^2 + (50 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$$

$$= \frac{1}{4} [64 + 0 + 100 + 36]$$

$$= 51$$

Variance Reduction

$$= \text{Var}(\text{Root}) - \sum w_i \text{Var}(\text{child})$$

$$= 60.8 - \left[\frac{1}{2} \times 100 + \frac{1}{2} \times 51 \right]$$

$$= 60.8 - 20 - 40.8$$

$$\boxed{\text{Var Red} = 0}$$

$$\text{Variance}(C_1) =$$

$$= \frac{1}{2} [(40 - 50)^2 + (42 - 50)^2]$$

$$= \frac{1}{2} [100 + 64]$$

$$= \frac{164}{2} = 82$$

$$\text{Variance}(C_2) =$$

$$= \frac{1}{3} [(52 - 50)^2 + (60 - 50)^2 + (56 - 50)^2]$$

$$= \frac{1}{3} [4 + 100 + 36]$$

$$= \frac{140}{3} = 46.66$$

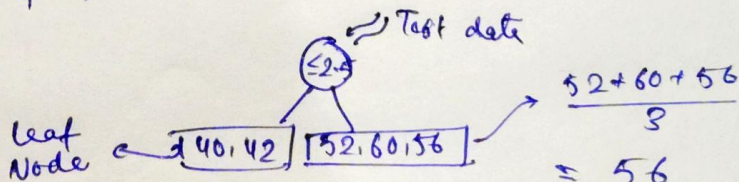
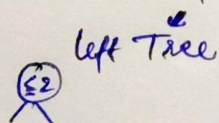
Variance Reduction

$$= 60.8 - \left[\frac{2}{5} \times 82 + \frac{3}{5} \times 46.66 \right]$$

$$= 60.8 - [32.8 + 27.996]$$

$$= +0.004$$

Variance Reduction (Left Split) < Variance Reduction (Right Split)



$$\text{o/p} = \frac{40 + 42}{2} = 41$$

o/p → Decision Tree Regressor